Integration by Substitution

Exam code: 9MA0

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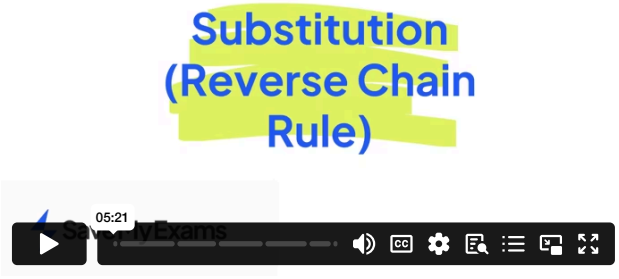


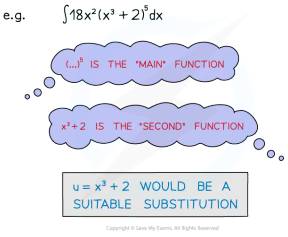
Did this video help you? Yes No

Help me understand

Integration by substitution

What is integration by substitution?

- Make sure you are familiar with Chain Rule and Reverse Chain Rule
- In more awkward problems it is harder to spot the reverse chain rule
- It is possible to use a substitution instead
- These kinds of substitutions usually will **not** be given
  - The substitution is deemed 'obvious'



How do I find a suitable substitution?

- Look to substitute the 'second' (rather than the 'main') function
- STEP 1: Determine the substitution

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

get rid of fractions)

- STEP 3: Replace all parts of the integral
  - all **x** terms should be replaced with equivalent **u** terms, including **dx**
  - if a definite integral change the limits from **x** to **u** too
- STEP 4: Integrate and either ...
  - ...substitute **x** back in
    or
  - ... evaluate the definite integral using the **u** limits
- STEP 5: Find **c**, the constant of integration, if needed

$I = \int 6u^5 du$

THE RESULT SHOULD BE REALLY EASY TO INTEGRATE

STEP 4: INTEGRATE, SUBSTITUTE x BACK IN

$I = u^6 + c$

$I = (x^3 + 2)^6 + c$

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FIND $\int_0^1 15x^2\cos(5x^3)dx$ GIVING YOUR ANSWER TO 3 SIGNIFICANT FIGURES.

$I = \int_0^1 15x^2\cos(5x^3)dx$

STEP 1: DETERMINE THE SUBSTITUTION

LET $u = 5x^3$

STEP 2: DIFFERENTIATE AND REARRANGE

$\frac{du}{dx} = 15x^2$

$du = 15x^2 dx$

STEP 3: REPLACE ALL PARTS OF THE INTEGRAL

$x = 0, \quad u = 0$

$x = 1, \quad u = 5$

CHANGE LIMITS TOO!

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$I = \int_0^5 \cos u\, du$

$\cos(5x^3) = \cos u$

$du = 15x^2 dx$

STEP 4: INTEGRATE AND EVALUATE

$I = [\sin u]_0^5$

$I = \sin 5 - \sin 0$

RADIANS FOR CALCULUS!

$I = -0.95892...$

$I = -0.959 \quad (3\text{ sf})$

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Examiner Tips and Tricks

- A lot of the "work" in these problems happens in your head :
  - what is the 'main' function?
  - what should the substitution be?
  - are there any numbers to 'adjust' and 'compensate' for?
- Be sure that what you write down is clear and easy to follow, and remember that you can check your final answer by differentiating it.

Worked Example

Given that y passes through the point $(1, 2e^3)$, and that

$$y = \int 5x^2 e^{x^3+2}\,\mathrm{d}x$$

find y in terms of x.

$y = \int 5x^2 e^{x^3+2}\,dx$

STEP 1: DETERMINE THE SUBSTITUTION

$e^{\cdots}$ IS THE MAIN FUNCTION

LET $u = x^3 + 2$

STEP 2: DIFFERENTIATE AND REARRANGE

$\frac{du}{dx} = 3x^2$

$du = 3x^2 dx$

STEP 3: REPLACE ALL PARTS OF THE INTEGRAL

$y = 5 \times \frac{1}{3}\int 3x^2 e^u\,dx$

CONSTANT

"ADJUST" AND "COMPENSATE"

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